

Deep Spatial–Spectral Representation Learning for Hyperspectral Image Denoising

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Abstract—Deep learning has found successful applications in restoration of two-dimensional (2-D) images including denoising, dehazing, and superresolution. However, existing deep convolutional neural network (DCNN) architecture cannot fully exploit spatial–spectral correlations in three-dimensional (3-D) hyperspectral images (HSIs) (directly extending 2-D DCNN into 3-D will significantly increase computational complexity); meantime, unlike 2-D images, there is an obstacle caused by the shortage of training data for HSIs. To meet those challenges, we present a novel, deep-learning framework for 3-D HSI denoising with the following contributions. First, inspired by the success of U-net in low-dose current-transformer denoising, we propose a novel approach of encoding rich multi-scale information of HSIs by a modified 3-D U-net. Second, we present a computationally efficient implementation of 3-D U-net based on the strategy of separable filtering. By decomposing 3-D filtering into 2-D spatial filtering and 1-D spectral filtering, we can achieve substantial savings on the number of network parameters to keep the computational complexity low. Third, we have developed a *transfer learning* approach of synthetically generating HSIs from RGB images as supplementary training data. The synthesized HSIs are used for the initial training of the modified 3-D U-net denoising network, which will be fine-tuned on real HSI images. Experimental results have shown that the proposed 3-D U-net denoising method significantly outperforms existing model-based HSI denoising methods.

Index Terms—Hyperspectral image denoising, modified 3-D U-net, separable filtering, spatial-spectral redundancy.

I. INTRODUCTION

HYPERSPECTRAL images (HSI) – images of the same scene captured across a number of different wavelengths – are important to various civilian and military applications including remote sensing [1], agriculture [2], object tracking [3] and recognition [4]. Existing hyperspectral imaging methods often suffer from the problem of low signal-to-noise ratio (SNR) due to the fundamental limitation in physics – i.e., simultaneously acquiring many images of different spectral bands inevitably

leads to shorter exposure time and narrower bandwidth. Accordingly, the acquired hyperspectral images are often corrupted by heavy noise which could severely deteriorate the performance of HSI-based segmentation and analysis algorithms. In view of the expensive cost of hardware-based solution to noise suppression, it is therefore desirable to develop effective software-based HSI denoising techniques as a preprocessing step to support various HSI-related applications.

Existing HSI denoising techniques in the literature can often be viewed as the extensions of grayscale image denoising methods. Early works including dictionary learning based methods [5], BM3D [6], and sparsity based denoising methods [7], [8] simply remove the noise in a band-by-band manner and cannot exploit strong correlation across the spectral bands. In order to exploit spatial-spectral correlations among HSI, a wavelet shrinkage method in the derivative-domain was proposed for HSI denoising in [9]; the principal component analysis (PCA) was combined with wavelet shrinkage methods in [10]; a spatial-spectral total variation model was developed for HSI denoising in [11]. This line of research has led to a multiple spectral-band conditional random fields (MSB-CRF) model [12] aiming at simultaneously characterizing the spatial-spectral dependencies in a unified probabilistic framework. Other notable works include a multidimensional Wiener filtering with adaptive tensor flattening [13], a vector formulation of bilateral filtering [14] and the extension of well-known BM3D method into BM4D [15].

More recently, the class of low-rank tensor approximation methods have been extended for HSI denoising [13], [16] as well. With Tucker decomposition, the multirank of HSI can be first estimated via the alternative least square (ALS) algorithm and then the noiseless HSI can be reconstructed from multirank truncation of the core tensor [13]. To address the uniqueness issue with Tucker decomposition during the estimation of multirank, the parallel factor analysis (PARAFAC) decomposition method has been proposed for HSI denoising in [17]. A nonlocal tensor dictionary learning method was proposed in [18] where similar noisy tensors grouped by similar patches are denoised by multirank truncation. Different from multirank truncation based tensor approximation, high-order singular value decomposition (HOSVD) is exploited in [19]. After HOSVD is applied to high-order tensors formed by similar patches, the core tensor coefficients were denoised via hard-thresholding followed by Wiener filtering. Most recently, Laplacian scale mixture (LSM) models were proposed for core tensor coefficients in [20]; the LSM

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prior led to an adaptive thresholding algorithm in which both coefficients and thresholds can be jointly estimated from noisy data.

Inspired by the success of deep convolutional neural networks (DCNN) on image classification [21], [22], DCNN has also been applied to various image restoration tasks – e.g., SRCNN [23], [24] and VDSR [25] for image super-resolution, Trainable nonlinear reaction diffusion (TNRD) [26] and DnCNN [27] for image denoising. In these methods, DCNNs are constructed to learn nonlinear mapping functions from the space of degraded images to that of target images (ground-truth) through end-to-end training. Due to the powerful capability of learning nonlinear mapping functions from large training data, DCNN-based approaches have achieved even better denoising results than conventional model-based denoising methods for grayscale images. To leverage the success of DCNN to HSI denoising, the original TNRD method [26] has been extended for HSI denoising in [28]. However, each spectral band image was still denoised separately in [28] and thus has not fully exploited the strong spectral correlations among HSI. Indeed, it is challenging to design *computationally efficient* DCNN to fully exploit the spatial-spectral correlations partially due to the high dimensionality of HSI data.

In this paper, we propose a novel approach of leveraging powerful representation learning capability of DCNN to the problem of HSI denoising. Note that directly extending 2D DCNN to 3D DCNN will significantly increase the number of parameters and therefore computational complexity. To meet this challenge, we propose a novel construction of 3D DCNN based on the idea of separable filtering – i.e., decomposing a 3D filtering into the concatenation of 2D spatial filtering and 1D spectral filtering. Inspired by the success of U-net [29], the spatial size of feature maps was reduced by introducing down-sampling layers to generate multi-scale spatial-spectral representation of noisy HSI data. Then the targeted spectral images can be incrementally reconstructed by fusing feature maps extracted from noisy HSI data across different scales, which fully exploits the rich multi-scale information of HSI. Furthermore, since it is difficult to obtain sufficient high-quality HSIs in practice, a special-purpose DCNN was designed to generate *synthetic* HSI data from natural RGB images. To the best of our knowledge, such DCNN-based data augmentation strategy has not been proposed for HSI in the open literature. Experimental results have shown that the proposed method can achieve better denoising performance than existing state-of-the-art HSI denoising methods including the latest Kronecker-basis-representation based tensor sparsity measure (KBR) [30]. The proposed 3D-DCNN has also been extended to deal with unknown noise levels and mixed noise types showing better results than other competing methods.

The rest of the paper is organized below. In Sec. II, we introduce the background and some related works. In Sec. III, we present the proposed 3D-DCNN for HSI denoising and elaborate the issues related to network architecture and end-to-end optimization. In Sec. V, we report the experimental results and compare them with several other competing HSI denoising methods. Sec. VI draws some conclusions of this paper.

II. RELATED WORKS

In this section, we briefly review previous works related to the proposed method – i.e., tensor-based HSI denoising methods, DCNN-based image denoising methods and existing 3D-DCNN for video recognition/analysis.

A. Tensor-Based HSI Denoising Methods

To exploit spatial-spectral correlation among HSI, tensor-based denoising methods have been developed as the extension of vector-based and matrix-based ones. In [13], [16], low-rank tensor approximation was applied to HSI where alternating least square (ALS) algorithm is used to estimate the multi-ranks of tensors. In [18], similar 3D patches were first grouped into high-order low-rank tensors and low-rank tensor approximation was then used to remove the noise. In view of the difficulty with multi-rank estimation, the PARAFAC decomposition model [17] and high-order SVD method [19] have also been proposed for low-rank tensor approximation. Instead of denoising the core tensors by hard-thresholding or Wiener filtering, a more principled approach based on Laplacian scale mixture model was developed in our own previous work [20]. Overall, all existing tensor-based HSI denoising methods consist of two key components: 1) Tucker or PARAFAC decomposition that transforms original tensors into low-rank representations; 2) a truncation or thresholding operation of core tensor coefficients to remove the noise. More recent low-rank based denoising for hyperspectral images include [31]–[34] and [35].

B. DCNN-Based Image Denoising Methods

In recent years, DCNNs have found successful application in natural image denoising and shown even better results than model-based methods (e.g., [26], [27], [36]). In [36], the multilayer perceptron (MLP) was applied to image denoising and achieved promising denoising performance that was comparable to the previous state-of-the-art BM3D [6]. Instead of stacking multiple neural network layers, an iterative nonlinear reaction diffusion model was unfolded into feed-forward networks in Trainable nonlinear reaction diffusion (TNRD) [26] whose parameters were optimized through end-to-end training. By stacking multiple convolutional layers without pooling layers and using a skip connection between the input and the output, a deep residue network implementation of DCNN has been proposed for image denoising in [27] leading to the current state-of-the-art denoising performance. Despite those advances, we note that the field of multi-dimensional image denoising has been under-researched. Among the few exceptions, [28] has extended the TNRD network into HSI denoising; however each spectral band image was still processed independently in [28] without exploiting strong spectral correlations within HSI. Most recently, deep neural networks have been proposed for HSI denoising in [37] and [38].

C. 3D DCNN for Video Analysis

DCNN and its 3D extensions have been proposed to exploit the spatial-temporal correlation for video analysis [39]–[44]. In

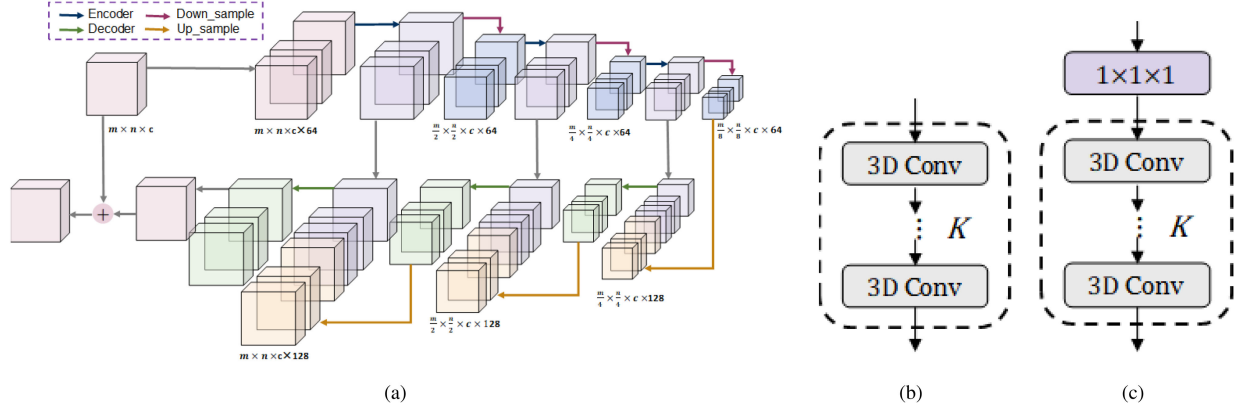


Fig. 1. Architecture of the proposed HSI denoising network. (a) The overall architecture of the proposed denoising network; (b)/(c) the encoding/decoding architecture of 3-D convolution building blocks.

[40], the spatial-temporal correlation of video was exploited by conducting spatial-temporal convolution over the stacked frame-level representation in a window of fixed size. Based on encoder and decoder of Long Short Term Memory (LSTM), the spatial-temporal representation of video was learned using an autoencoder model in [41]. To directly learn the spatial-temporal representation over consecutive frames, 3D-DCNN has been proposed in [39]; even deeper 3D-DCNNs have been proposed to learn the spatial-temporal representation from longer video clips in [42], [44]. However, it is often difficult to train a deep 3D-CNN due to prohibitive computational complexity and memory demand. To address this issue, [43] proposed to replace the 3D convolution by a 2D spatial convolution and 1D temporal convolution leading to significant reduction in terms of model size and computational complexity. Inspired by this work, we propose to decompose the 3D convolution into 2D spatial convolution and 1D *spectral* convolution. Different from [43], our approach integrates the separable 3D convolution into an autoencoder and decoder denoising framework in order to more effectively exploit the multi-scale redundancies of HSI as we will elaborate next.

III. PROPOSED 3-D CONVOLUTIONAL NETWORK FOR HSI DENOISING

In this section, we first present the overall 3D convolutional network as a general framework of HSI processing and then describe the details of 3D convolutional layer used for the denoising application.

A. Network Architecture

The overall architecture of the proposed HSI denoising network is shown in Fig. 1(a). Inspired by the U-net for semantical segmentation [29] and sharpMask network for object segmentation [45], the proposed network contains feature encoding and decoding components as part of nonlinear mapping. The input noisy HSI of size $m \times m \times s$ first goes through a serial of feature encoding stages to generate embedded spatial-spectral features; those features are then taken as the input to the decoding component to reconstruct a clean HSI. There are L ($L = 3$ in

our implementation) encoding stages to generate a multi-scale spatial-spectral representation of the input HSI. As shown in Fig. 1(b), each encoding stage contains K ($K = 5$) 3D convolutional layers, followed by a downsampling layer to reduce the spatial resolution of feature maps by a scaling factor of 2 (the pooling layer helps to increase the receptive field of individual neurons). Specifically, a 3×3 filter with stride 2 is applied to the feature maps to reduce the spatial resolution by a scaling factor of 2. Thus, the feature maps extracted at the l -th ($l = 1, 2, \dots, L$) encoding stage is of dimension $\frac{m}{2^l} \times \frac{m}{2^l} \times s \times N$, where N ($N = 64$ in our implementation) denotes the number of separable 3D convolutional kernels. The details of the separable 3D convolutional will be given in the next subsection.

In parallel, feature decoding stages progressively reconstruct the HSI from encoded feature maps through L decoding stages. Since the finally extracted feature maps have a coarse spatial resolution, it is often difficult to recover fine image details from coarse-resolution feature maps. To remedy the loss of spatial resolution, feature maps at the same spatial resolution extracted during the encoding stages are combined together with those upsampled feature maps. As shown in Fig. 1(c), the feature maps of a larger size are concatenated with those upsampled feature maps, followed by a convolutional layer with a $1 \times 1 \times 1$ kernel to reduce the dimensions. We note that such a bottleneck design has been widely used in modern deep neural networks for reducing the overall computational complexity. Similar to the encoding stages, we have adopted 5 convolutional layers with separable 3D convolutional kernels and also added a skip connection from the input noisy HSI to the reconstructed HSI to enforce the network to learn the residual instead of the original HSI (similar idea has been adopted in the previous work of DnCNN [27]).

B. Separable 3-D Convolution

For a given HSI patch of size $m \times m \times s$, a 3D convolutional filter of size $k \times k \times s$ can be used to exploit the spatial-spectral correlation. However, direct implementation of 3D convolution results in a large number of parameters and therefore prohibitive computational complexity. Inspired by the previous work [43],

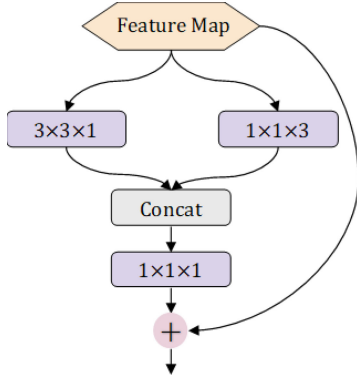


Fig. 2. The implementation of 3-D convolution building block via separating into 2-D spatial filter and the 1-D spectral filter.

TABLE I
SEPARABLE 3-D CONV VS. 3-D CONV FOR I.I.D. GAUSSIAN DENOISING ON CAVE DATASET

	Noise level	PSNR	SSIM	ERGAS	SAM
Separable 3D Conv	$\sigma=30$	40.71	0.9639	47.92	0.1211
3D Conv		39.50	0.9475	54.97	0.1694

we propose to decompose the 3D convolution filter (e.g., $3 \times 3 \times 3$) into a 2D spatial filter of size $3 \times 3 \times 1$ and a 1D spectral filter of size $1 \times 1 \times 3$. As shown in Fig. 2, the input is filtered with the 2D spatial filter and the 1D spectral filter in a parallel fashion. The outputs of the two filtered feature maps are concatenated to obtain the desirable result. Such separable 3D convolution can significantly reduce the number of parameters. To further reduce the dimension of concatenated feature maps, a $1 \times 1 \times 1$ convolutional layer is added. Meantime, similar to the existing deep residual network, a skip connection from the input to the output is also included.

To justify the efficiency and effectiveness of the proposed strategy, we have compared the performance of 3D convolution with and without separable implementation on CAVE dataset in Table I. In addition to conventional metrics such as PSNR and SSIM, we have included two objective metrics specially tailored for HSI: relative dimensionless global error in synthesis (ERGAS) [46] and spectral angle mapper (SAM) [47]. It can be observed that separable implementation outperforms its counterpart for all considered quality metrics (note that for the latter two error-related evaluation terms, lower values imply better qualities). Some exemplar visual quality comparison can be found in Fig. 3.

C. Network Training

The overall deep convolutional network shown in Fig. 1 has been optimized by end-to-end training. The mean square error (MSE) based loss function was employed to train the proposed deep network as follows

$$\Theta = \underset{\Theta}{\operatorname{argmin}} \sum_{i=1}^N \|\mathcal{F}(\mathbf{y}_i; \Theta) - \mathbf{x}_i\|_F^2, \quad (1)$$

where $\mathbf{y}_i$ and $\mathbf{x}_i$ denote the i -th pair of noisy and original spectral image patches respectively, and $\mathcal{F}(\cdot; \Theta)$ denotes the reconstructed spectral image patch by the network using the parameter set Θ . The ADAM optimizer [48] was used to train the network with parameter setting $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. The learning rate is initialized as 10^{-3} . The proposed network was implemented using the TensorFlow framework and trained with one Nvidia Titan XP GPU.

IV. SYNTHETIC TRAINING DATA: TRANSFER LEARNING FROM RGB IMAGES

Unlike RGB images whose training data are widely available on the web (e.g., ImageNet), HSI data are often prioritized and only some small dataset has been made publicly accessible. Therefore, there is an obstacle of training data shortage in the field of HSI denoising. Similar problems also existing in the fields of single image dehazing [49] and underwater image enhancement [50] where real-world training data are difficult to acquire. To overcome such difficulty, model-based or learning-based (e.g., via generative adversarial network [51]) have been proposed to simulate large-scale training data on computers. Along this line of reasoning, it is natural to explore the possibility of synthesizing HSI data from RGB images.

There have been some attempts to generate spectral images from RGB images [52], [53]. In [52], a radial basis function network has been proposed to reconstruct a scene's spectral reflectance from a single RGB image. Specifically, mapping functions between RGB pixel and a spectral pixel were learned. However, this method ignored the correlations between neighboring pixels. Recently, Fu *et al.* proposed to recover the spectral reflectance from a single RGB image using non-negative sparse representation [53]. Through dictionaries learning and structured sparse representation, promising spectral recovery results have been achieved. However, the sparse coding based method fail to jointly optimize the dictionaries and the regularization parameters.

Unlike previous approaches, we propose a novel approach of recovering the spectral reflectance from a single RGB image using deep convolutional neural networks (DCNN) as shown in Fig. 4. In our approach, we have used a small number of HSI data and their corresponding RGB images from CAVE and HARVARD dataset respectively as the training data. The constructed RGB-to-HSI network has seven 2d-convolution layers, and except the last layer each layer is followed by a ReLU activation layer. After training, we use 300 images from BSD dataset of natural scenes as the test set to generate the corresponding synthetic HSI images. We have experimentally verified that the synthetic HSI images have rich spatial information such as edges and textures, which justifies the effectiveness of the proposed DCNN for RGB-to-HSI conversion. We use these synthetic HSI images to pre-train our denoising network and then use the public available real HSI datasets to fine-tune the network parameters.

The transfer learning convolutional network shown in Fig. 4 has been optimized by end-to-end training. The mean square error (MSE) based loss function was employed to train the

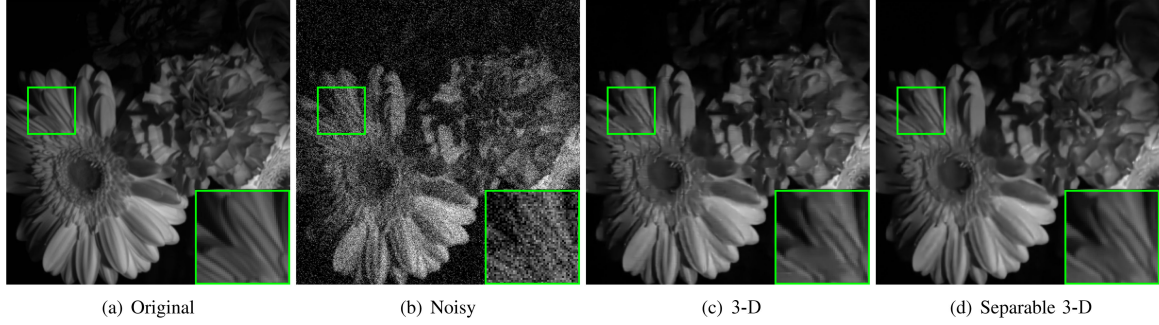


Fig. 3. i.i.d. gaussian denoising results for *flowers* image on band 26 with noise level 30. (a) Original HSI; (b) Noisy HSI (20.45 dB); (c) 3-D Conv (38.70 dB); (d) Separable 3-D (38.89 dB).

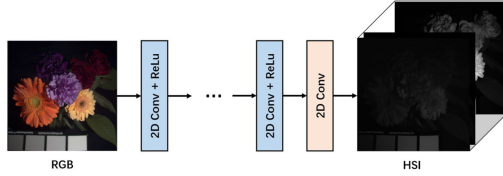


Fig. 4. The proposed network for generating HSI data from RGB ones.

TABLE II
GENERATE HSIs QUALITY ON CAVE DATASET

PSNR	SSIM	ERGAS	SAM
35.03	0.9569	109.71	0.1871

proposed deep network as follows

$$\Theta = \underset{\Theta}{\operatorname{argmin}} \sum_{i=1}^N \|\mathcal{F}(\mathbf{y}_i; \Theta) - \mathbf{x}_i\|_F^2, \quad (2)$$

where $\mathbf{y}_i$ and $\mathbf{x}_i$ denote the i -th pair of original spectral image patches and corresponding RGB image patches respectively, and $\mathcal{F}(\cdot; \Theta)$ denotes the reconstructed spectral image patch by the network using the parameter set Θ . The ADAM optimizer [48] was used to train the network with the same parameter setting as before ($\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$). The batch size is set as 64. The learning rate is initialized as 10^{-3} and train about 50 epochs. The proposed network was implemented using the TensorFlow framework and trained with one Nvidia Titan XP GPU. As some preliminary evidence for the quality of synthesized HSI by the proposed transfer learning, we have reported the evaluation on CAVE dataset using the four objective quality metrics (same as before) in Table II.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

In this section, we report our experimental results of several HSI denoising tasks to demonstrate the performance of the proposed network. To verify the performance of the proposed network for HSI denoising, the commonly used CAVE [54] and Harvard [55] datasets have been used for both training and testing. The CAVE dataset consist of 32 hyperspectral images under daily life scenes; the images of size $512 \times 512 \times 31$ are captured

with the wavelengths in the range of 400–700 nm at the interval of 10 nm. The Harvard dataset contains 50 hyperspectral images in real indoor and outdoor scenes; the spatial resolution of each image is 1040×1392 ; the spectral band range is 420–720 nm (the band interval is still 10 nm). Specifically, the central part of size 512×512 is cropped from each image of Harvard dataset for testing and training. We randomly selected 18 spectral images from the CAVE dataset and 20 images from the Harvard dataset as training data respectively, and use the remaining 12 images of CAVE dataset and 30 images of the Harvard dataset as test data.

We have set the patch size to be $50 \times 50 \times 10$ and obtained around 110,000 patches (after patch augmentation with flips and rotations) to train the model. In our experiments, all HSI data are prescaled to the interval of $[0, 1]$. During the training, the loss function is chosen to be the average mean squared error between the desired residual images and estimated ones from noisy input. We use Adam (adaptive moment estimation) with a batch size of 16. The initial learning rate is set as $\beta_1 = 0.9$ and the total number of epochs is 20–50. We initialize the weights by the random normal method with a standard deviation of 0.02 and train different denoising networks for both i.i.d. Gaussian noise and non-i.i.d. Gaussian noise in our experiments. Our implementation is based on Tensorflow; and all our experiments are run on a workstation with one NVIDIA Titan X GPU. We have implemented two different versions: *Ours* – 1 is trained using public HSI images only; while *Ours* – 2 is first trained using synthetic HSI images (i.e., to pre-train our denoising network) and then fine-tuned by public HSI datasets.

To simulate the real-world noise for HSIs, several different types of noise are used to simulate the noisy HSIs, including Gaussian noise, Stripe noise and impulse noise. Specifically, the following three types of noise are added to the clean HSIs to generate noisy HSIs:

- 1) i.i.d Gaussian noise, i.e., each spectral band was corrupted by i.i.d Gaussian noise with noise level $\sigma = 10, 20$, or 30;
- 2) Non-i.i.d Gaussian noise, i.e., each spectral band was corrupted by Gaussian noise with different noise levels such that the SNR of each band is generated from uniform distribution in the range of $[10, 30]$ dB;
- 3) Non-i.i.d Gaussian + Stripe noise, i.e., all the bands were first corrupted by the non-i.i.d Gaussian noise as described

TABLE III
QUANTITATIVE RESULTS OF COMPETING METHODS FOR I.I.D. GAUSSIAN DENOISING ON CAVE DATASET (THE BEST RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noise level	PARAFAC	LRTA	BM4D	TDL	KBR	NAILRMA	FastHyDe	LLRT	Ours-1	Ours-2
PSNR	$\sigma=10$	32.72	40.38	43.92	43.75	44.47	40.54	43.07	46.08	45.61	45.73
SSIM		0.8364	0.9431	0.9783	0.9794	0.9789	0.9288	0.9783	0.9871	0.9845	0.9850
ERGAS		194.05	49.67	32.69	33.39	31.15	49.19	36.93	25.57	27.3	26.75
SAM		0.2714	0.1766	0.1217	0.1001	0.1346	0.2229	0.1260	0.0670	0.0940	0.0890
PSNR	$\sigma=20$	32.49	37.09	40.25	39.95	41.17	36.04	39.72	42.53	42.23	42.56
SSIM		0.8231	0.8993	0.9532	0.9556	0.9591	0.8593	0.9562	0.9754	0.9719	0.9730
ERGAS		125.92	72.11	49.94	51.53	45.41	81.04	54.85	38.58	40.61	38.76
SAM		0.3031	0.2167	0.1903	0.1531	0.2018	0.3529	0.2019	0.0908	0.1071	0.1119
PSNR	$\sigma=30$	32.26	35.07	38.03	37.64	39.46	33.36	37.63	40.55	40.15	40.71
SSIM		0.8081	0.8601	0.9247	0.9281	0.9434	0.7629	0.9284	0.9650	0.9596	0.9640
ERGAS		128.99	90.86	64.60	67.06	55.12	108.98	70.43	49.50	50.95	47.92
SAM		0.3237	0.2464	0.2479	0.2006	0.2333	0.4626	0.2779	0.1096	0.1288	0.1211

TABLE IV
QUANTITATIVE RESULTS OF COMPETING METHODS FOR I.I.D. GAUSSIAN DENOISING ON HARVARD DATASET (THE BEST RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noise level	PARAFAC	LRTA	BM4D	TDL	KBR	NAILRMA	FastHyDe	LLRT	Ours-1	Ours-2
PSNR	$\sigma=10$	35.39	42.80	43.68	45.02	45.05	42.42	45.6	46.12	45.75	45.95
SSIM		0.8737	0.9552	0.9423	0.9769	0.9762	0.9492	0.9811	0.9813	0.9804	0.9813
ERGAS		108.35	45.22	40.29	33.85	33.90	46.05	32.16	29.87	31.17	30.28
SAM		0.1202	0.0831	0.0762	0.0581	0.0629	0.0709	0.0638	0.0498	0.0571	0.0574
PSNR	$\sigma=20$	35.19	39.97	40.34	41.92	42.17	38.02	43.07	43.21	42.05	43.39
SSIM		0.8690	0.9249	0.9441	0.9489	0.9621	0.8672	0.9681	0.9659	0.9563	0.9687
ERGAS		110.54	61.97	60.31	48.65	44.40	77.82	43.17	43.02	48.90	41.11
SAM		0.1249	0.0926	0.1052	0.0760	0.0728	0.1263	0.0875	0.0607	0.0765	0.0676
PSNR	$\sigma=30$	35.17	38.21	38.34	39.80	41.08	35.28	41.38	41.43	40.61	41.98
SSIM		0.8522	0.8976	0.9171	0.9336	0.9478	0.7731	0.9552	0.9522	0.9476	0.9581
ERGAS		110.60	77.68	77.18	61.15	54.00	107.62	52.58	55.39	56.49	48.61
SAM		0.1280	1008	0.1300	0.0964	0.0821	0.1960	0.1132	0.0693	0.0778	0.0717

in case (2) and then the Stripe noise was then added to 8 randomly selected bands. The number of stripes in each band is from 20–40;

- 4) Non-i.i.d Gaussian noise + impulse noise, i.e., all the bands were first corrupted by the non-i.i.d Gaussian noise as described in case (2) and then the impulse noise with different intensity was added to 8 randomly selected bands. The percentage of impulse is from 50% to 70%.

B. Objective Performance Comparison

We have compared the PSNR/SSIM/ERGAS/SAM results of the proposed methods (*Ours* – 1 and *Ours* – 2) against a dozen existing competing methods – i.e., PARAFAC [17], lower rank tensor approximation (LRTA) [56], low-rank matrix recovery (LRMR) [57], LLRT [34], Total-variation-regularized low-rank matrix factorization (LRTV) [58] (including its enhanced version LLRSSTV [33]), BM4D [15] (with default cube size),¹ tensor dictionary learning (TDL) [18], KBR [30], NAILRMA [31], FastHyDe [35] and the non i.i.d. mixture of Gaussian based method (NMoG) [59]. Note that both the PSNR and SSIM values were calculated over all bands between clean HSI and denoised HSI. The experimental results of all competing methods on Harvard and CAVE HSI data are shown in Tables III–VI respectively.

¹We have tested that other cube sizes such as $8 \times 8 \times 8$, $4 \times 4 \times 16$ and $4 \times 4 \times 32$ lead to slightly better or even worse performance.

It can be seen that 1) pretraining does improve the denoising performance (*Ours* – 2 uniformly outperforms *Ours* – 1); 2) the proposed method consistently outperforms all other competing methods on both i.i.d. noise and non-i.i.d. noise cases. On i.i.d. Gaussian noise case, from these quantitative comparison, the PSNR gain of the proposed method over other competing methods can be easily observed – the average PSNR improvements over BM4D, TDL, and KBR methods are as large as 2.06 dB, 1.43 dB and 0.80 dB respectively. The performance improvement in the situation of non-i.i.d. noise cases is even more impressive – the average PSNR improvements over LRMR, LRTV, BM4D, and NMoG methods are as much as 8.87 dB, 8.13 dB, 4.17 dB and 6.43 dB, respectively. It should be noted that some benchmark method such as BM4D is not designed for HSI data but general 3D video frames. We also note that those benchmark methods are mostly designed with the assumption of i.i.d. noise and their performance degrades rapidly in the presence of non-i.i.d. noise; by contrast, our proposed technique is a lot more robust to the variations of noise characteristics.

To further demonstrate the generalization capability of the proposed denoising network, we have tested it on a dataset (DC Mall) acquired with different spectral range (covering 400–2400 nm). This dataset is acquired by the hyperspectral digital imagery collection experiment (HYDICE) sensor and its original size is $1280 \times 307 \times 191$ (we have cropped two separated subsets each sized by $200 \times 200 \times 191$: one used as training and the other for testing). Tables VII and VIII have included

TABLE V
QUANTITATIVE RESULTS OF COMPETING METHODS FOR NON-I.I.D. GAUSSIAN DENOISING ON CAVE DATASET (THE TOP TWO RANKING RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noisy HSI	LRMR	LRTV	BM4D	NMoG	NAILRMA	FastHyDe	LLRSSTV	Ours-1	Ours-2
Non-i.i.d. Gaussian Noise										
PSNR	34.93	41.62	41.11	45.27	42.05	43.76	46.47	41.83	49.20	49.60
SSIM	0.7255	0.9212	0.9729	0.9622	0.9558	0.9534	0.9890	0.9578	0.9919	0.9922
ERGAS	131.86	54.31	47.64	35.78	55.33	47.88	26.38	54.07	17.69	16.89
SAM	0.4149	0.2701	0.1222	0.1796	0.2296	0.2244	0.1064	0.2089	0.0787	0.0778
Non-i.i.d. Gaussian + Stripe Noise										
PSNR	32.70	39.63	40.87	41.14	40.83	40.48	44.61	40.84	49.08	49.44
SSIM	0.7255	0.9039	0.9702	0.9168	0.9491	0.9369	0.9793	0.9554	0.9916	0.9919
ERGAS	167.09	73.54	49.88	99.24	68.38	69.76	32.54	62.50	17.86	17.14
SAM	0.4534	0.3030	0.1384	0.2389	0.2438	0.2564	0.1274	0.2282	0.0784	0.0775
Non-i.i.d. Gaussian + Impulse Noise										
PSNR	32.51	38.66	40.25	41.47	39.84	40.39	44.93	40.05	48.83	48.95
SSIM	0.6844	0.9072	0.9676	0.9452	0.9407	0.9492	0.9851	0.9543	0.9912	0.9916
ERGAS	202.61	86.70	53.15	90.20	87.34	69.25	32.38	73.58	18.51	18.13
SAM	0.4541	0.2793	0.1336	0.1757	0.2520	0.2110	0.1178	0.2152	0.0774	0.0779

TABLE VI
QUANTITATIVE RESULTS OF COMPETING METHODS FOR NON-I.I.D. GAUSSIAN DENOISING ON HARVARD DATASET (THE TOP TWO RANKING RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noisy HSI	LRMR	LRTV	BM4D	NMoG	NAILRMA	FastHyDe	LLRSSTV	Ours-1	Ours-2
Non-i.i.d. Gaussian Noise										
PSNR	40.20	44.58	44.88	47.48	46.88	46.69	49.31	45.95	49.33	49.46
SSIM	0.8331	0.9426	0.9658	0.9776	0.9679	0.9780	0.9901	0.9704	0.9899	0.9901
ERGAS	95.20	48.67	39.77	28.85	41.89	30.65	20.91	40.49	20.59	20.27
SAM	0.2702	0.1445	0.0942	0.0835	0.1119	0.0775	0.0507	0.1052	0.0482	0.0479
Non-i.i.d. Gaussian + Stripe Noise										
PSNR	37.92	42.16	44.46	43.46	46.55	44.11	48.10	44.43	49.28	49.43
SSIM	0.7972	0.9266	0.9605	0.9314	0.9731	0.9674	0.9837	0.9680	0.9898	0.9903
ERGAS	142.60	79.12	47.92	104.56	50.63	51.36	28.28	55.54	20.73	20.38
SAM	0.3071	0.1636	0.1062	0.1558	0.0988	0.0965	0.0639	0.1060	0.0481	0.0475
Non-i.i.d. Gaussian + Impulse Noise										
PSNR	37.38	41.91	44.17	44.73	45.72	44.72	48.78	44.40	48.81	48.98
SSIM	0.7635	0.9214	0.9598	0.9583	0.9620	0.9712	0.9893	0.9627	0.9889	0.9893
ERGAS	157.92	73.88	45.73	53.37	66.06	39.97	22.77	59.95	22.05	21.56
SAM	0.3350	0.1565	0.1011	0.0978	0.1246	0.0804	0.0530	0.1229	0.0494	0.0479

TABLE VII
QUANTITATIVE RESULTS OF COMPETING METHODS FOR I.I.D. GAUSSIAN DENOISING ON WASHINGTON DC MALL DATASET (THE BEST RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noise level	PARAFAC	LRTA	BM4D	TDL	KBR	NAILRMA	FastHyDe	Ours-1	Ours-2
PSNR	$\sigma=30$	40.99	45.76	40.97	41.34	44.38	37.93	44.37	42.74	45.79
SSIM		0.9117	0.9391	0.9381	0.9337	0.9546	0.8966	0.9690	0.9483	0.9590
ERGAS		32.48	24.28	24.29	24.04	19.83	29.07	15.97	21.29	18.45
SAM		0.0195	0.0147	0.0174	0.0158	0.0132	0.0177	0.0102	0.0142	0.0118

TABLE VIII
QUANTITATIVE RESULTS OF COMPETING METHODS FOR NON-I.I.D. GAUSSIAN DENOISING ON WASHINGTON DC MALL DATASET (THE BEST RESULTS IN EACH ROW ARE HIGHLIGHTED BY BOLDFACE)

	Noisy HSI	LRMR	LRTV	BM4D	NMoG	NAILRMA	FastHyDe	LLRSSTV	Ours-1	Ours-2
PSNR	22.73	38.37	45.84	36.81	45.03	40.11	47.74	41.81	45.28	48.88
SSIM	0.2751	0.9147	0.9547	0.7784	0.9774	0.9414	0.9882	0.9477	0.9779	0.9842
ERGAS	265.16	29.14	22.69	74.28	15.52	25.87	12.90	47.72	14.19	11.55
SAM	0.2224	0.0215	0.0155	0.0198	0.0626	0.0116	0.0193	0.0378	0.0106	0.0086

the objective performance comparison results for iid and non-iid Gaussian denoising experiments (ours-1: trained on CAVE; ours-2: fine-tuned on synthetically generated HSIs). It can be observed that the proposed network still consistently outperform other competing approaches. Note that even for model-based

approaches, there is a potential model-data mismatch which could severely degrade the performance on dataset beyond the reach of adopted models. Our experimental results have clearly justified that the proposed denoising network can be generalized to other dataset (different from the training ones).

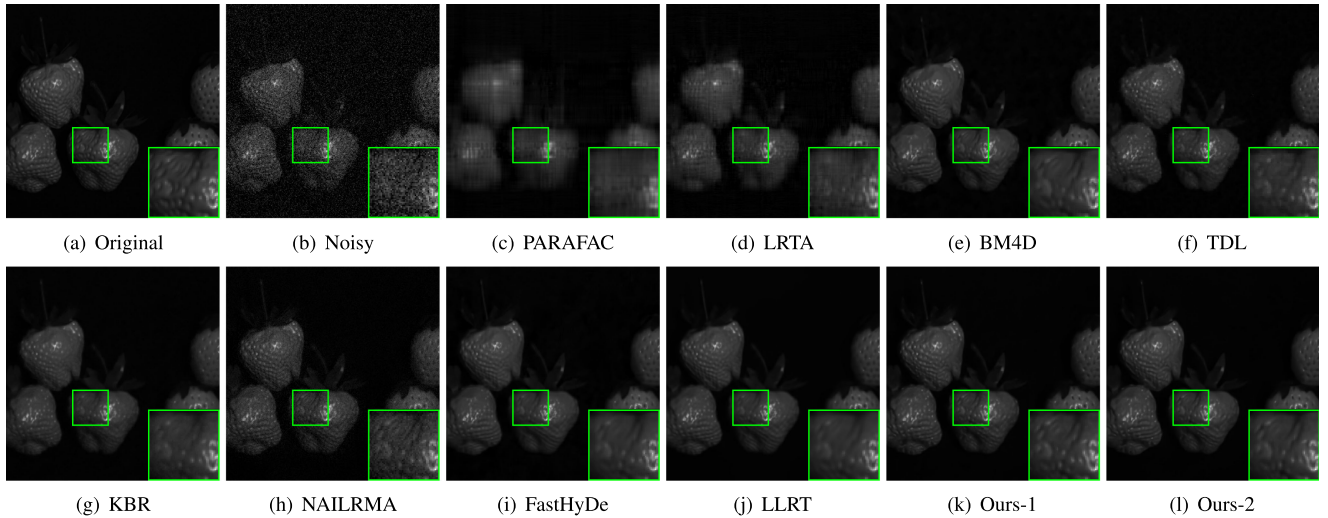


Fig. 5. i.i.d. gaussian denoising results for *fake and real strawberries* image on band 27 with noise level 20. (a) Original HSI; (b) Noisy HSI (23.92 dB); (c) PARAFAC [60] (34.44 dB); (d) LRTA [56] (37.21 dB); (e) BM4D [61] (41.32 dB); (f) TDL [62] (40.90 dB); (g) KBR [30] (41.98 dB); (h) NAILRMA [31] (36.35 dB); (i) FastHyDe [35] (41.61 dB); (j) LLRT [34] (42.86 dB); (k) Ours-1 (**43.33** dB); (l) Ours-2 (**43.27** dB).

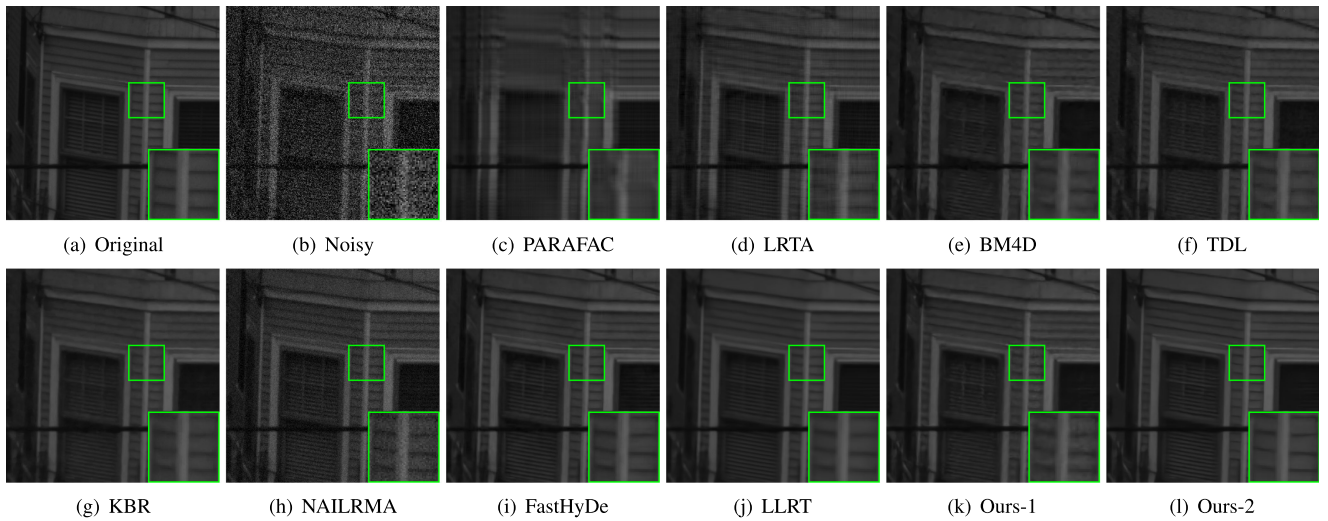


Fig. 6. i.i.d. gaussian denoising results for *c4* image on band 23 with noise level 30. (a) Original HSI; (b) Noisy HSI (19.67 dB); (c) PARAFAC [60] (31.15 dB); (d) LRTA [56] (35.85 dB); (e) BM4D [61] (37.00 dB); (f) TDL [62] (38.22 dB); (g) KBR [30] (39.20 dB); (h) NAILRMA [31] (31.13 dB); (i) FastHyDe [35] (39.85 dB); (j) LLRT [34] (39.89 dB); (k) Ours-1 (38.23 dB); (l) Ours-2 (**40.83** dB).

C. Subjective Performance Comparison

We have also compared the subjective quality of denoised images among ours (*Ours-1* and *Ours-2*) and eight other competing approaches: PARAFAC [60], LRTA [56], BM4D [61], TDL [62], KBR [30], NAILRMA [31], FastHyDe [35] and LLRT [34]. In all testing scenarios, we have observed that *Ours-2* (with pre-training) can restore richer spatial information than *Ours-1* (without pre-training). Fig. 5–12 contains the comparison of denoised images for four different noise settings, as we will elaborate next.

Fig. 5 and Fig. 6 show the visual comparison of different denoising methods for i.i.d. Gaussian noise with $\sigma = 20$. It can

be seen that the images restored by our method are capable of better preserving fine image structures while removing Gaussian noise. We note that benchmark methods such as PARAFAC, LRTA and BM4D tend to produce over-smoothed edges and textures; TDL and KBR perform slightly better while our method apparently recovers more details such as the texture patterns of *strawberries* in Fig. 5 and clear wall lines and metal frames of *c4* in Fig. 7.

Fig. 7 and Fig. 8 compare the restoration result of various competing methods for non-i.i.d. Gaussian noise. It can be seen that competing method such as LRMR performs poorly on removing non-i.i.d. Gaussian noise; LRTV can remove non-i.i.d. Gaussian noise but at the price of image blur. Similarly, BM4D

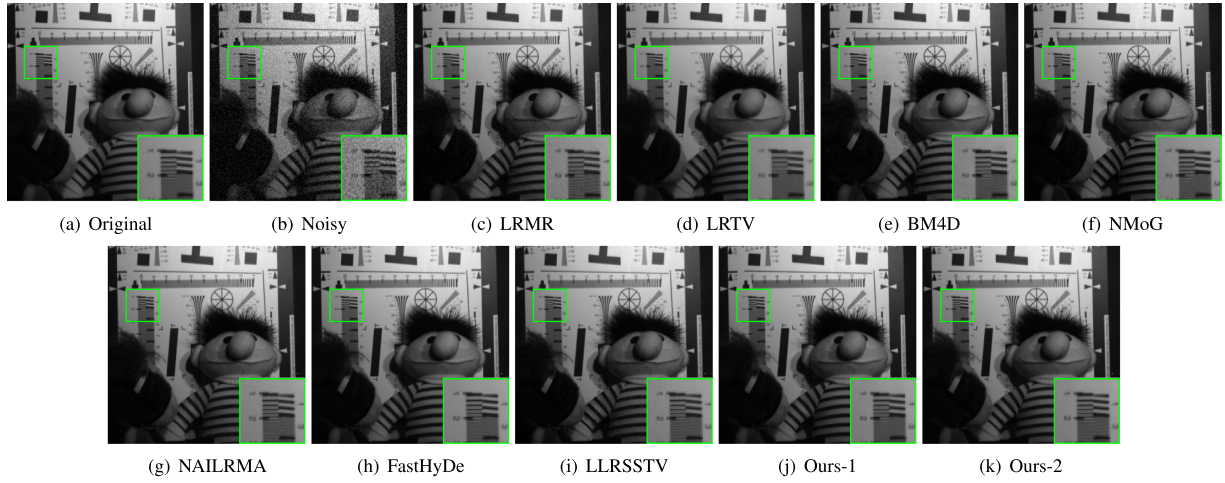


Fig. 7. Non-i.i.d. gaussian denoising results for *chart* and *stuffed toy* image on band 28. (a) Original HSI; (b) Noisy HSI (26.10 dB); (c) LRMR [57] (37.04 dB); (d) LRTV [58] (39.64 dB); (e) BM4D [61] (39.11 dB); (f) NMoG [59] (38.27 dB); (g) NAILRMA [31] (40.56 dB); (h) FastHyDe [35] (44.06 dB); (i) LLRSSTV [33] (38.87 dB); (j) Ours-1 (**45.23** dB); (k) Ours-2 (**45.79** dB).

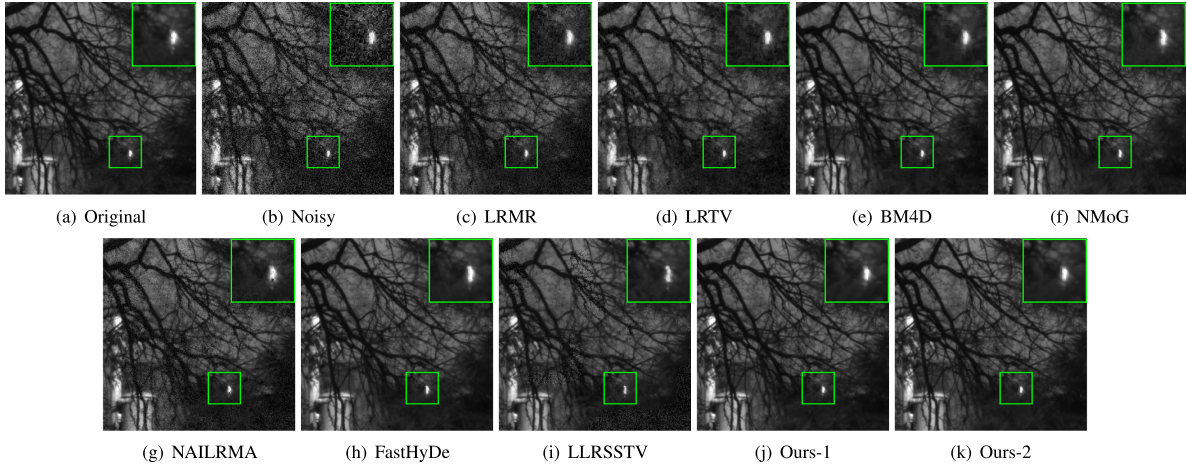


Fig. 8. Non-i.i.d. gaussian denoising results for *b8* image on band 25. (a) Original HSI; (b) Noisy HSI (20.58 dB); (c) LRMR [57] (25.15 dB); (d) LRTV [58] (28.26 dB); (e) BM4D [61] (34.16 dB); (f) NMoG [59] (37.88 dB); (g) NAILRMA [31] (25.73 dB); (h) FastHyDe [35] (39.88 dB); (i) LLRSSTV [33] (25.04 dB); (j) Ours-1 (**40.33** dB); (k) Ours-2 (**41.40** dB).

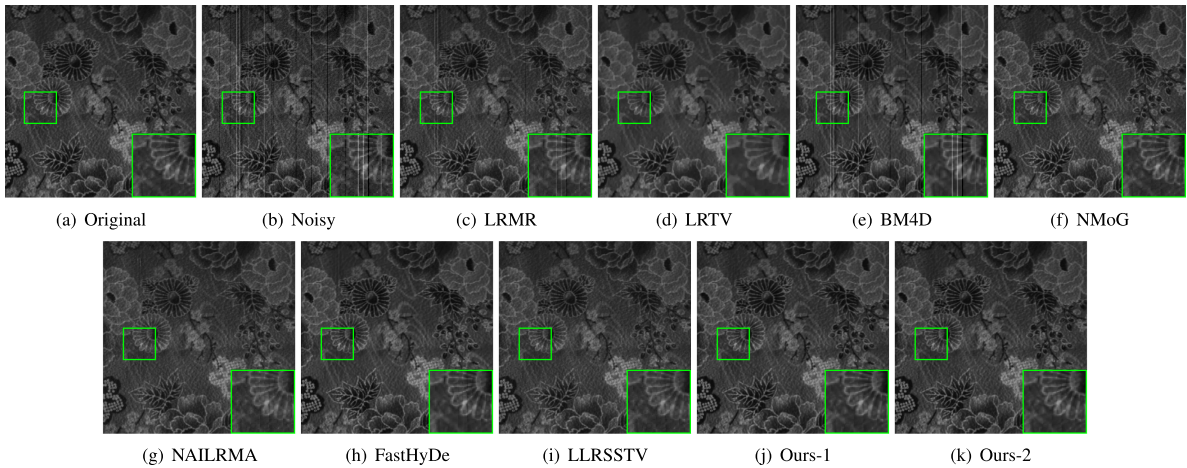


Fig. 9. Non-i.i.d. gaussian with stripe denoising results for *cloth* image on band 25. (a) Original HSI; (b) Noisy HSI (28.03 dB); (c) LRMR [57] (36.35 dB); (d) LRTV [58] (34.42 dB); (e) BM4D [61] (30.78 dB); (f) NMoG [59] (40.30 dB); (g) NAILRMA [31] (35.04 dB); (h) FastHyDe [35] (39.34 dB); (i) LLRSSTV [33] (36.11 dB); (j) Ours-1 (**41.78** dB); (k) Ours-2 (**42.00** dB).

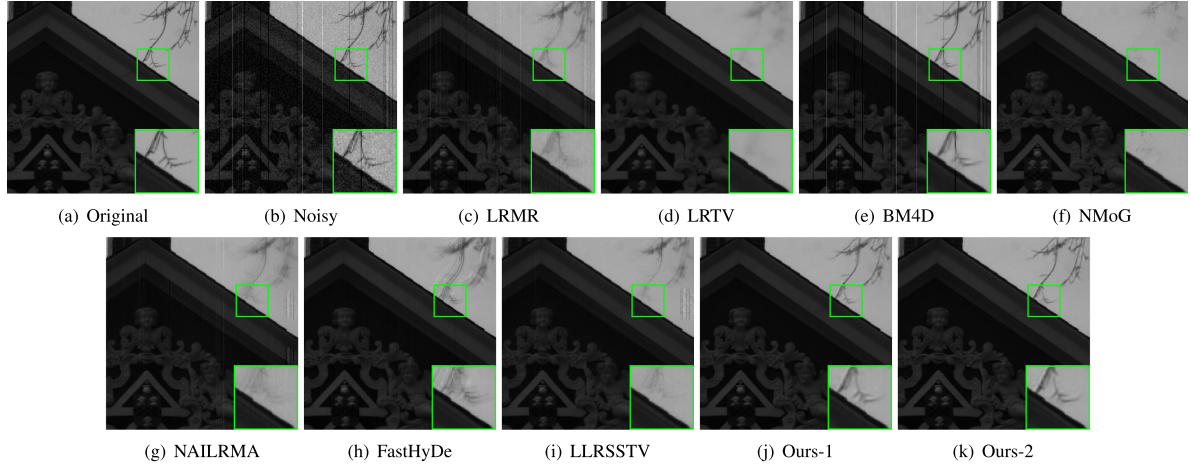


Fig. 10. Non-i.i.d. gaussian with stripe denoising results for *e1* image on band 18. (a) Original HSI; (b) Noisy HSI (26.12 dB); (c) LRM [57] (34.32 dB); (d) LRTV [58] (36.21 dB); (e) BM4D [61] (30.56 dB); (f) NMoG [59] (35.25 dB); (g) NAILRMA [31] (34.82 dB); (h) FastHyDe [35] (36.82 dB); (i) LLRSSTV [33] (35.34 dB); (j) Ours-1 (**40.75 dB**); (k) Ours-2 (**41.40 dB**).

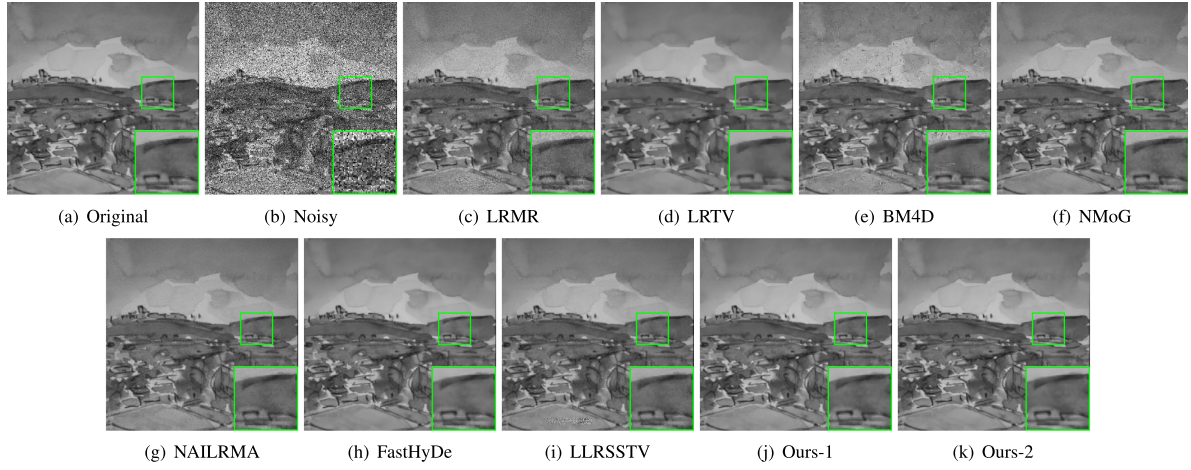


Fig. 11. Non-i.i.d. gaussian with impulse denoising results for *watercolor* image on band 24. (a) Original HSI; (b) Noisy HSI (16.90 dB); (c) LRM [57] (30.45 dB); (d) LRTV [58] (35.90 dB); (e) BM4D [61] (31.39 dB); (f) NMoG [59] (37.36 dB); (g) NAILRMA [31] (35.20 dB); (h) FastHyDe [35] (40.11 dB); (i) LLRSSTV [33] (35.11 dB); (j) Ours-1 (40.05 dB); (k) Ours-2 (**41.05 dB**).

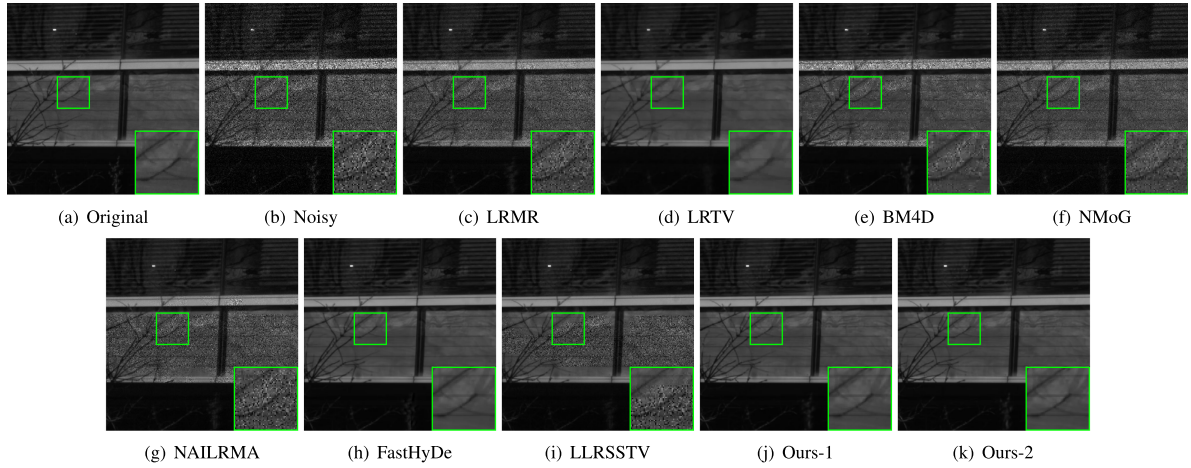


Fig. 12. Non-i.i.d. gaussian with impulse denoising results for *f3* image on band 24. (a) Original HSI; (b) Noisy HSI (23.15 dB); (c) LRM [57] (28.50 dB); (d) LRTV [58] (38.46 dB); (e) BM4D [61] (28.64 dB); (f) NMoG [59] (28.83 dB); (g) NAILRMA [31] (28.13 dB); (h) FastHyDe [35] (40.30 dB); (i) LLRSSTV [33] (35.11 dB); (j) Ours-1 (**42.64 dB**); (k) Ours-2 (**43.26 dB**).

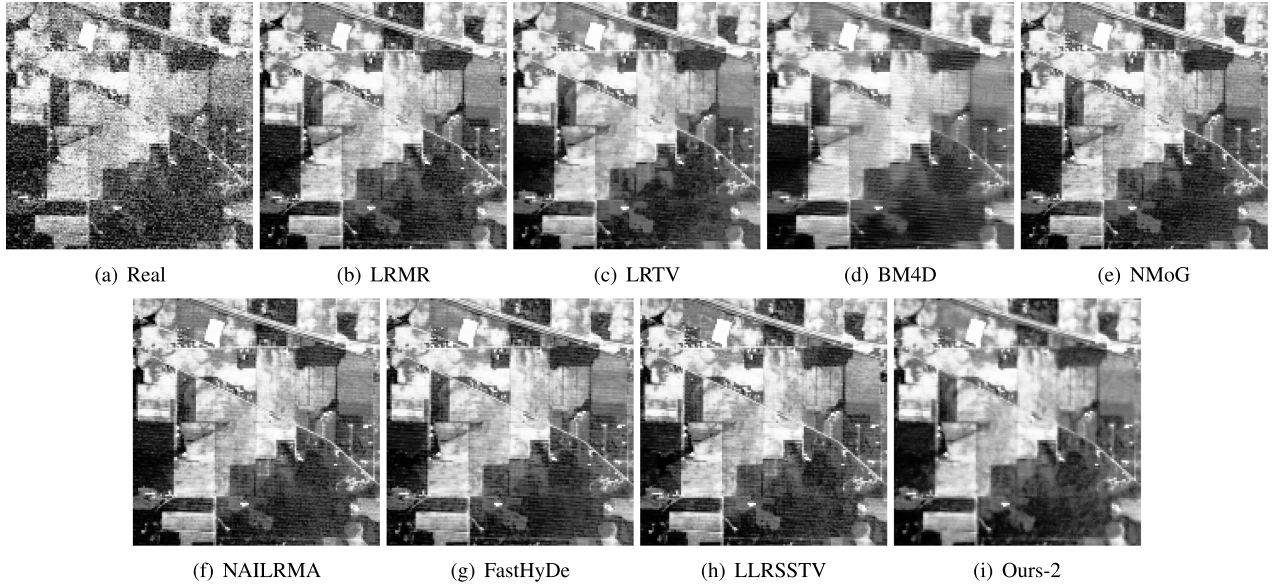


Fig. 13. Denoising results for *Indian Pines* on band 3. (a) Real HSI; (b) LRMR [57]; (c) LRTV [58]; (d) BM4D [61]; (e) NMoG [59]; (f) NAILRMA [31]; (g) FastHyDe [35]; (h) LLRSSTV [33]; (i) Ours-2.

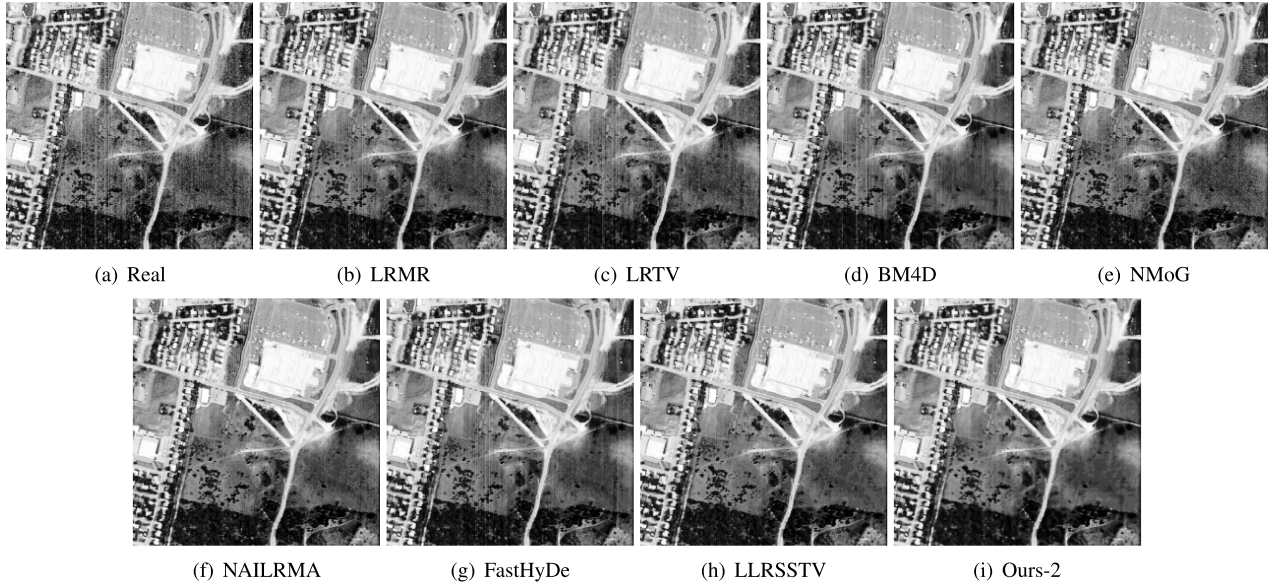


Fig. 14. Denoising results for *Urban* on band 2. (a) Real HSI; (b) LRMR [57]; (c) LRTV [58]; (d) BM4D [61]; (e) NMoG [59]; (f) NAILRMA [31]; (g) FastHyDe [35]; (h) LLRSSTV [33]; (i) Ours-2.

can not completely remove non-i.i.d. Gaussian noise such as the blur of *chart and stuffed toy* in Fig. 7. NMoG's performance is inconsistent – it can remove non-i.i.d. Gaussian noise and get some blur such as tree branches of *b8* in Fig. 8; but it seems doesn't work well such as the band 31 of *chart and stuffed toy* in Fig. 7 (this is the reason why NMoG does not perform well on the quantitative evaluation).

For Non-i.i.d. Gaussian with stripe or impulse noise, Fig. 9–Fig. 12 show that LRMR can remove mixed noise slightly, LRTV restored HSI to some degree but blur many details, BM4D can remove non-i.i.d. Gaussian noise but can hardly remove

stripe noise, and NMoG can remove mixed noise but suffer from loss of important image structures. By contrast, our approach is capable of achieving the best tradeoff between preserving image details and removing unwanted noise (including strip and impulse noise). Visual quality improvements over other competing methods can be easily justified in Fig. 10 and Fig. 12.

Finally, we want to demonstrate the performance of the proposed denoising network on real-world data. Figs. 13 and 14 include the denoising results comparison for two real-world images: *Indian Pines* and *Urban*. Due to lack of ground-truth, it is more difficult to tell which denoising technique delivers the

most faithful reproduction; but visual inspection alone seems to suggest that the proposed denoising network has achieved at least comparable performance to other leading methods such as LLRSSTV. In order to facilitate visual inspection, we have applied standard histogram equalization techniques to all images (so their contrast can be enhanced).

VI. CONCLUSION

Deep learning based approach towards HSI denoising has two technical obstacles: the computational bottleneck with 3D-DCNN implementation and the shortage of high-quality training data. To meet those challenges, we present a novel deep learning framework for 3D hyperspectral image denoising with a modified 3D U-net admitting a computationally efficient separable filtering implementation and a computer-based training data simulation via transfer learning. Inspired by the success of U-net, we propose a novel approach of encoding rich multi-scale information of hyperspectral images by a modified 3D U-net. To achieve a computationally efficient implementation of 3D U-net, we propose to decompose 3D filtering into 2D spatial filtering and 1D spectral filtering, which can achieve substantial savings on the number of network parameters. Additionally, we propose to synthetically generate hyperspectral images from RGB images using a novel DCNN. The synthesized hyperspectral images are used for initially training 3D U-net denoising network which will be fine tuned on real HSI images. Experimental results show that the proposed 3D U-net denoising method significantly outperforms existing state-of-the-art (model-based) HSI denoising methods.

REFERENCES

- [1] J. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.
- [2] Y. Tarabalka, J. Chanussot, and J. A. Benediktsson, "Segmentation and classification of hyperspectral images using minimum spanning forest grown from automatically selected markers," *IEEE Trans. Syst., Man, Cybern., Syst. Part B (Cybern.)*, vol. 40, no. 5, pp. 1267–1279, Oct. 2010.
- [3] H. V. Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2010, pp. 41–51.
- [4] M. Uzair, A. Mahmood, and A. Mian, "Hyperspectral face recognition using 3d-DCT and partial least squares," in *Proc. Brit. Mach. Vis. Conf.*, 2013, pp. 1–57.
- [5] M. Elad and M. Aharon, "Image denoising via sparse and redundant representation over learned dictionaries," *IEEE Trans. Image Process.*, vol. 15, no. 12, pp. 3736–3745, Dec. 2006.
- [6] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-d transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.
- [7] W. Dong, L. Zhang, G. Shi, and X. Li, "Nonlocally centralized sparse representation for image restoration," *IEEE Trans. Image Process.*, vol. 22, no. 4, pp. 1620–1630, Apr. 2013.
- [8] W. Dong, G. Shi, Y. Ma, and X. Li, "Image restoration via simultaneous sparse coding: Where structured sparsity meets Gaussian scale mixture," *Int. J. Comput. Vis.*, vol. 114, pp. 217–232, 2015.
- [9] H. Othman and S.-E. Qian, "Noise reduction of hyperspectral imagery using hybrid spatial-spectral derivative-domain wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 2, pp. 397–408, Feb. 2006.
- [10] G. Chen and S. Qian, "Denoising of hyperspectral imagery using principal component analysis and wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 973–980, Mar. 2011.
- [11] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral-spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.
- [12] P. Zhong and R. Wang, "Multiple-spectral-band CRFs for denoising junk bands of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2260–2275, Apr. 2013.
- [13] D. Letexier and S. Bourennane, "Noise removal from hyperspectral images by multidimensional filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2061–2069, Jul. 2008.
- [14] H. Peng, R. Rao, and S. A. Dianat, "Multispectral image denoising with optimized vector bilateral filter," *IEEE Trans. Image Process.*, vol. 23, no. 1, pp. 264–273, Jan. 2014.
- [15] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.
- [16] A. Karami, M. Yazdi, and A. Z. Asli, "Noise reduction of hyperspectral images using kernel non-negative Tucker decomposition," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 487–493, Jun. 2011.
- [17] X. Liu, S. Bourennane, and C. Fossati, "Denoising of hyperspectral images using the PARAFAC model and statistical performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3717–3724, Oct. 2012.
- [18] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 2949–2957.
- [19] A. Rajwade, A. Rangarajan, and A. Banerjee, "Image denoising using the higher order singular value decomposition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 4, pp. 849–862, Apr. 2013.
- [20] W. Dong, G. Li, G. Shi, X. Li, and Y. Ma, "Low-rank tensor approximation with Laplacian scale mixture modeling for multiframe image denoising," in *Proc. IEEE Conf. Comput. Vis.*, 2015, pp. 442–449.
- [21] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
- [22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [23] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2014, pp. 184–199.
- [24] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2016, pp. 391–407.
- [25] J. Kim, J. Lee, and K. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1646–1654.
- [26] Y. Chen and T. Pock, "Trainable nonlinear reaction diffusion: A flexible framework for fast and effective image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1256–1272, Jun. 2017.
- [27] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.
- [28] W. Xie and Y. Li, "Hyperspectral imagery denoising by deep learning with trainable nonlinearity function," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1963–1967, Nov. 2017.
- [29] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervention*, 2015, pp. 234–241.
- [30] Q. Xie, Q. Zhao, D. Meng, and Z. Xu, "Kronecker-basis-representation based tensor sparsity and its applications to tensor recovery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 8, pp. 1888–1902, Aug. 2018.
- [31] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.
- [32] L. Zhuang and J. M. Bioucas-Dias, "Hyperspectral image denoising based on global and non-local low-rank factorizations," in *Proc. IEEE Int. Conf. Image Process.*, 2017, pp. 1900–1904, pp. 2672–2680.
- [33] W. He, H. Zhang, H. Shen, and L. Zhang, "Hyperspectral image denoising using local low-rank matrix recovery and global spatial-spectral total variation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 713–729, Mar. 2018.
- [34] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. Comput. Vis. Pattern Recognit.*, 2017, pp. 5901–5909.

- [35] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.
- [36] H. C. Burger, C. J. Schuler, and S. Harmeling, "Image denoising: Can plain neural networks compete with BM3D?," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 2393–2399.
- [37] Y. Chang, L. Yan, H. Fang, S. Zhong, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 99, pp. 667–682, Feb. 2018.
- [38] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial-spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.
- [39] S. Ji, W. Xu, M. Yang, and K. Yu, "3d convolutional neural networks for human action recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 221–231, Jan. 2013.
- [40] A. Karpathy, G. Toderici, S. Shetty, T. Leung, R. Sukthankar, and L. Fei-Fei, "Large-scale video classification with convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 1725–1732.
- [41] N. Srivastava, E. Mansimov, and R. Salakhutdinov, "Unsupervised learning of video representations using LSTMs," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 843–852.
- [42] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3d convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4489–4497.
- [43] Z. Qiu, T. Yao, and T. Mei, "Learning spatio-temporal representation with pseudo-3d residual networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 5533–5541.
- [44] G. Varol, I. Laptev, and C. Schmid, "Long-term temporal convolutions for action recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 6, pp. 1510–1517, Jun. 2018.
- [45] P. O. Pinheiro, T.-Y. Lin, R. Collobert, and P. Dollar, "Learning to refine object segments," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 75–91.
- [46] L. Alparone, S. Baronti, A. Garzelli, and F. Nencini, "A global quality measurement of pan-sharpened multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 313–317, Oct. 2004.
- [47] R. H. Yuhas, A. F. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. JPL, Summaries 3rd Annu. Airborne Geosci. Workshop*, 1992, pp. 147–149.
- [48] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent.*, 2014.
- [49] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, "Dehazenet: An end-to-end system for single image haze removal," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5187–5198, Nov. 2016.
- [50] C. Fabbri, M. Jahidul Islam, and J. Sattar, "Enhancing underwater imagery using generative adversarial networks," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2018, pp. 7159–7165.
- [51] I. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
- [52] R. M. H. Nguyen, D. K. Prasad, and M. S. Brown, "Training-based spectral reconstruction from a single RGB image," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 186–201.
- [53] Y. Fu, Y. Zheng, L. Zhang, and H. Huang, "Spectral reflectance recovery from a single RGB image," *IEEE Trans. Comput. Imag.*, vol. 4, no. 3, pp. 382–394, Sep. 2018.
- [54] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Post-capture control of resolution, dynamic range and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.
- [55] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2011, pp. 193–200.
- [56] N. Renard, S. Bourennane, and J. Blanc-Talon, "Denoising and dimensionality reduction using multilinear tools for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 138–142, Apr. 2008.
- [57] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014.
- [58] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.
- [59] Y. Chen, X. Cao, Q. Zhao, D. Meng, and Z. Xu, "Denoising hyperspectral image with non-i.i.d. noise structure," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 1054–1066, Mar. 2018.
- [60] X. Liu, S. Bourennane, and C. Fossati, "Denoising of hyperspectral images using the PARAFAC model and statistical performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3717–3724, Oct. 2012.
- [61] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–113, Jan. 2013.
- [62] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. Comput. Vis. Pattern Recognit.*, 2014, pp. 2949–2956.



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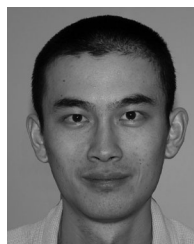


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